Research

I study the use of data in education, especially in STEM education. My research spans two interconnected lines of work: using new data sources and computational methods to describe and question educational systems, and weaving data science into teaching and learning, with a focus on science education.

In the first line of work, I study educational systems using computational approaches and novel data sources. This includes research on educators' professional learning through social media, critical investigations of how K-12 institutions' social media practices compromise student privacy, and methodological work that integrates machine learning with qualitative analysis.

In the second line of work, I develop and study approaches for data science learning in K-12 settings. This includes designing instruction for statistical reasoning — especially Bayesian thinking — in science classrooms, building tools and curricula that make data science accessible to newcomers, and contributing to the development of educational data science as a field through open-source software, a co-authored book (*Data Science in Education Using R*, Routledge, 2020), and a synthesis of what is known about K-12 data science learning.

Across both lines, I program in R and Python, use advanced statistical methods (especially multilevel models), develop open-source tools, and center partnerships with teachers. My current and near-future work extends these efforts into multimodal learning analytics, community-based data science learning, and the roles of generative AI in education.

Links

- Publications
- Google Scholar profile
- Project CREDIBLE - NSF CAREER-funded project
- Data Exploration and Sensemaking - NSF-funded collaborative project
- Making Data Science Count - research group
- *Data Science in Education Using R* - open-access version of the book

- CV (PDF)